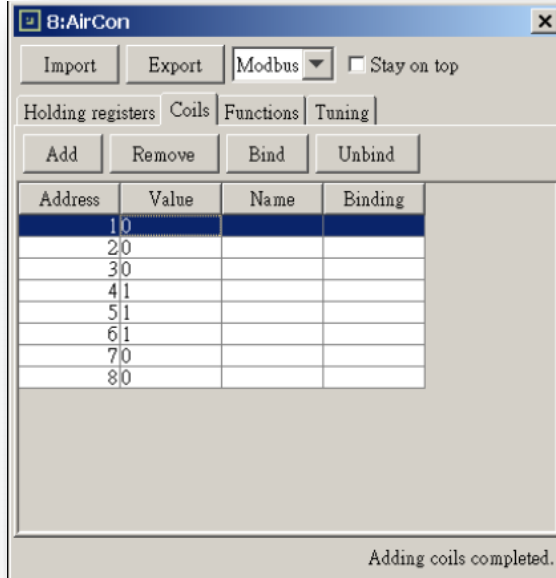
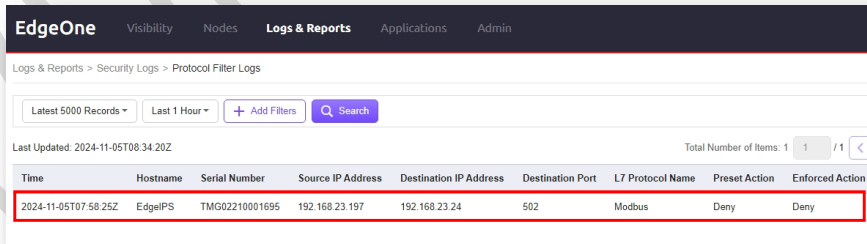
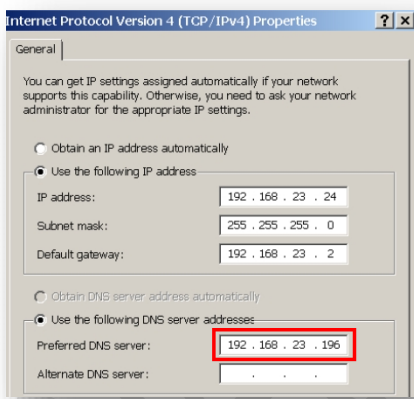
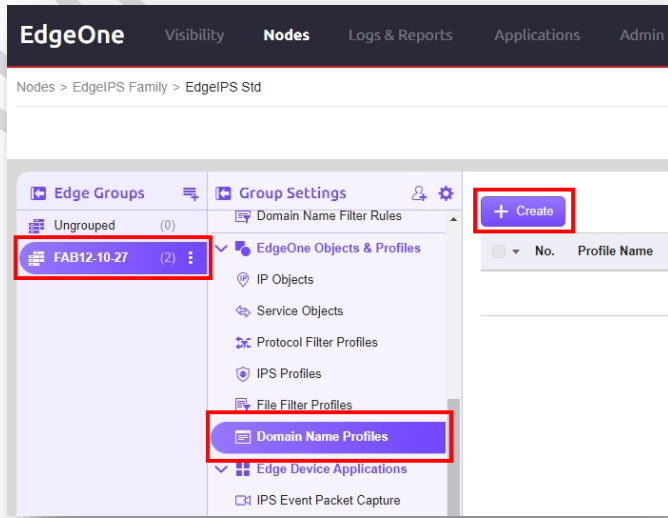
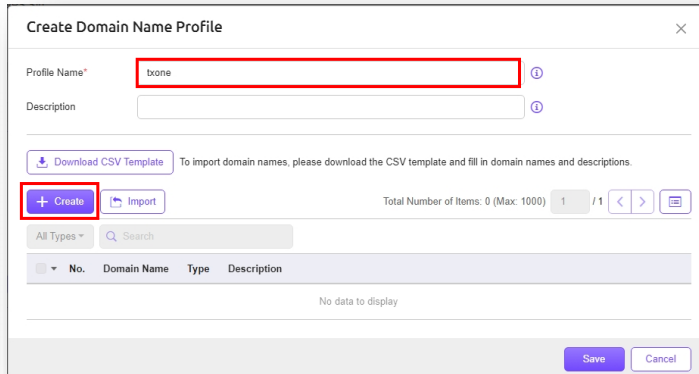
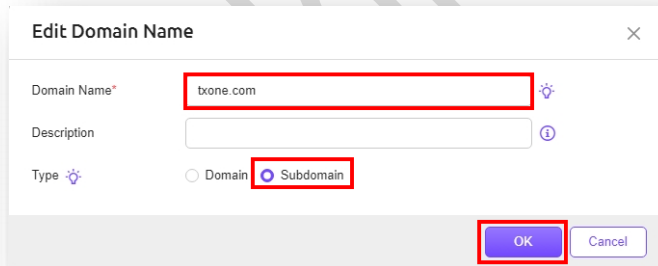
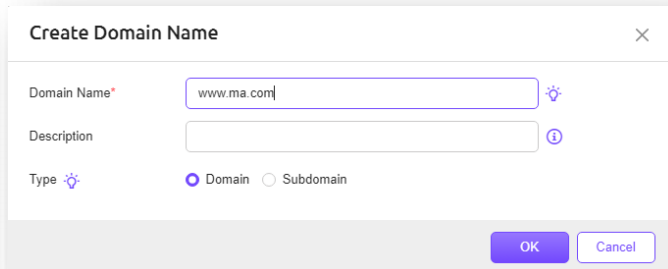


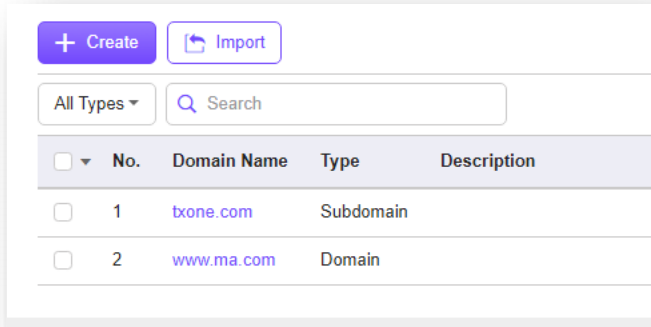
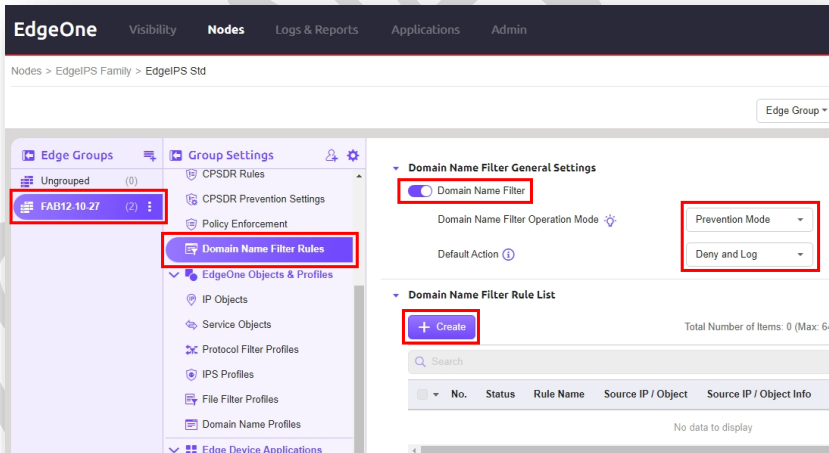
Items	Descriptions
	4. Check the Modbus slave simulator. The Address 1 value that has not changed and remained at “0”  5. Navigate to the EdgeOne web UI. 6. Go to the “Logs > Protocol Filter Logs” page. 7. Verify the logs events that blocked unauthorized behaviors. 
Expected Result	EdgeIPS can successfully block unauthorized commands.

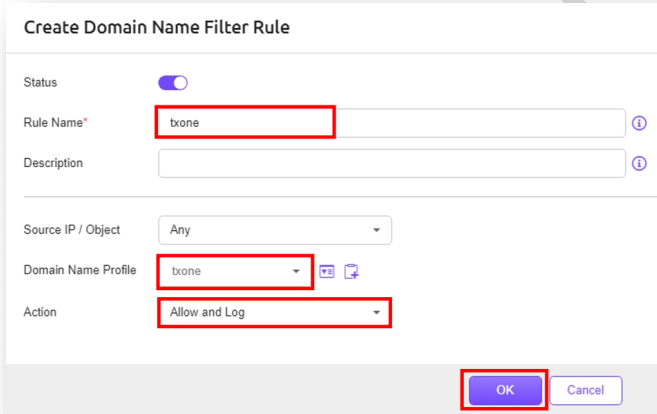
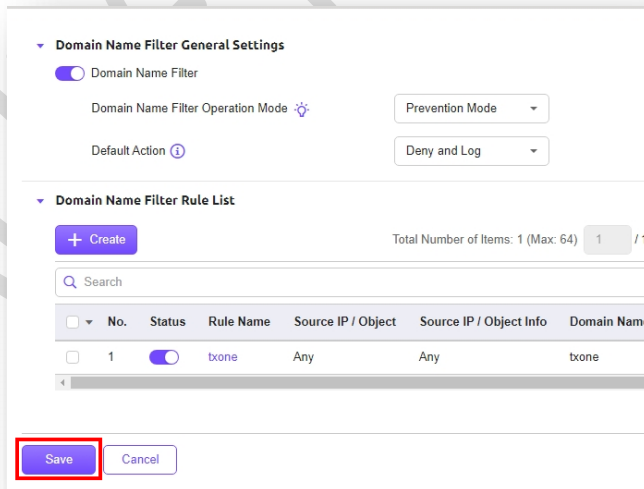
Lab6: EdgeIPS Domain Name Filter

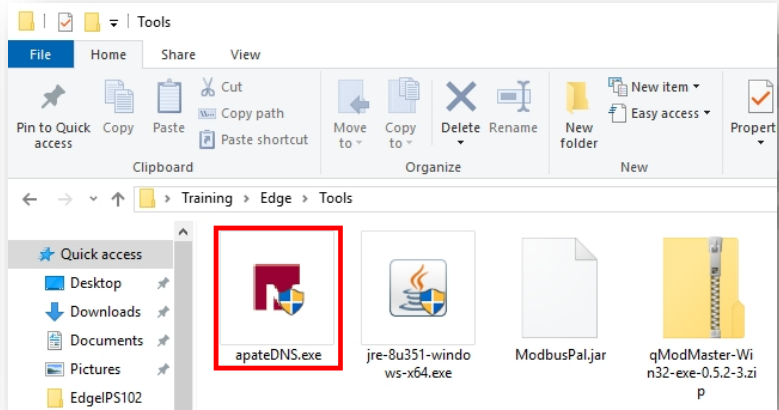
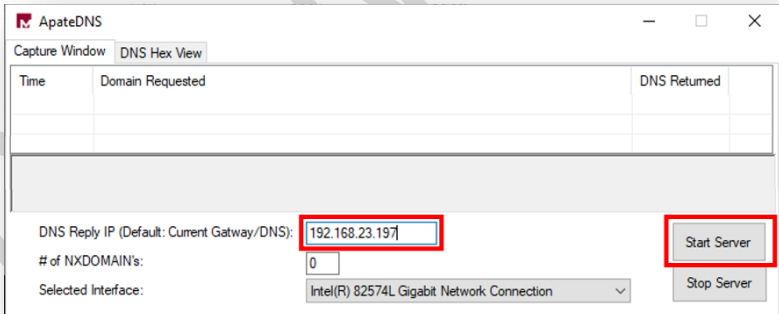
Table 9 EdgeIPS Domain Name Filter

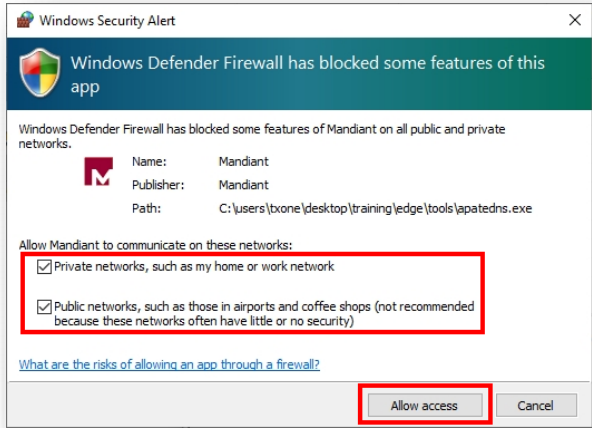
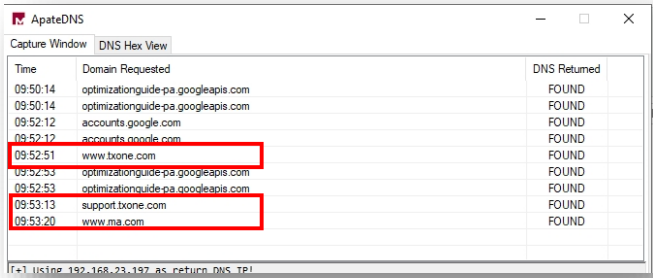
Items	Descriptions
Lab Case	Allow authorized DNS queries to pass through the EdgeIPS
Steps	- Log on to the Win7 VM and configure the DNS resolver to point to the LabConsole.  - Navigate to the EdgeOne web UI. - Go to Node >EdgeIPS Family > EdgeIPS Std > Edge Group “FAB12-20-27” \ Domain Name Profiles and click Create. 

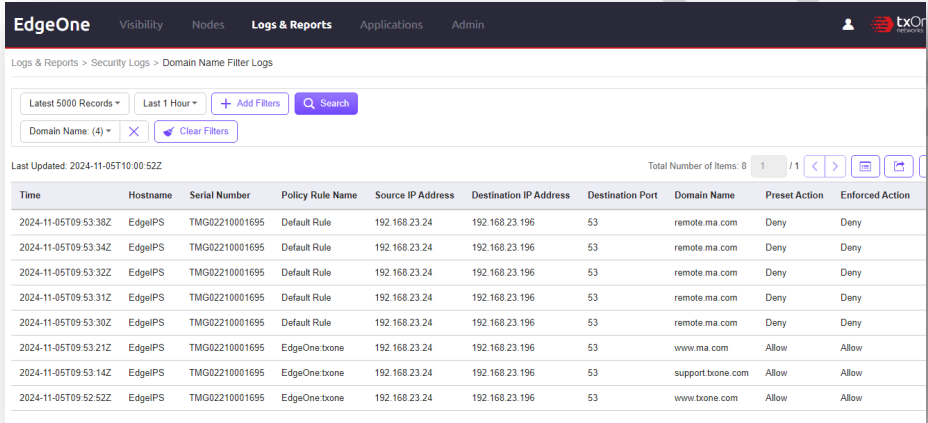
Items	Descriptions
	4. Naming the Profile Name “txone” and click Create.  5. Enter the domain name information and click OK. - Domain Name: txone - Type: Subdomain  6. Enter another Domain Name information. - Domain Name: www.ma.com - Type: Domain 

Items	Descriptions
	7. There are now two domain name records in the profile. Click Save.  8. Go to Domain Name Filter Rules. - Enable the "Domain Name Filter" - Switch the "Domain Name Filter Operation Mode" to Prevention Mode - Click Create. 

Items	Descriptions
	9. Create the Domain Name Filter Rule. - Rule Name: txone - Domain Name Profile: select “txone” - Action: Allow and Log - Click OK  10. Click Save. 

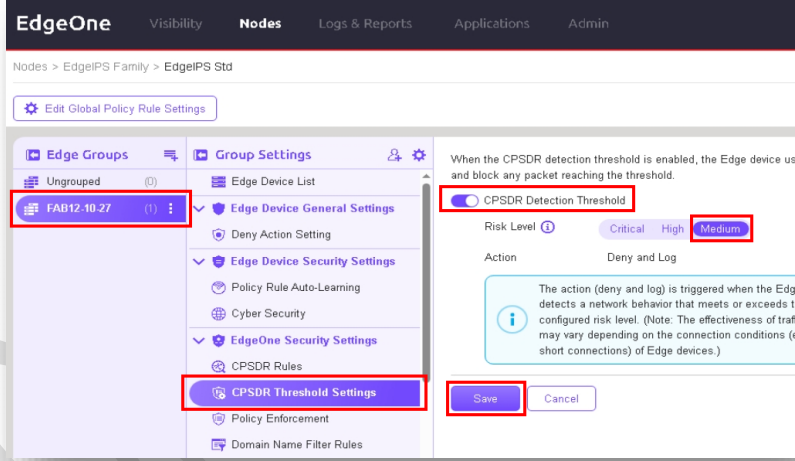
Items	Descriptions
	Test the Domain Name Filter Rules 1. Log on to the LabConsole 2. Access to the folder C:\Users\txone\Desktop\Training\Edge\Tools 3. Run the tool apateDNS.exe to start the local DNS Server  4. Configure the DNS Reply IP: 192.168.23.197 and click Start Server. 

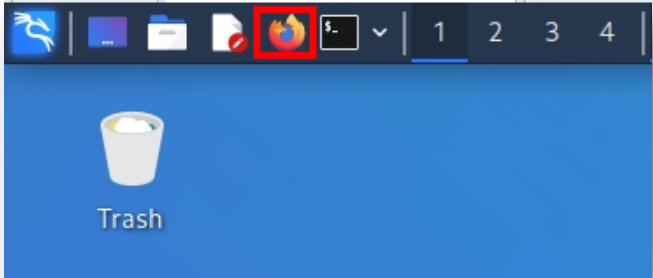
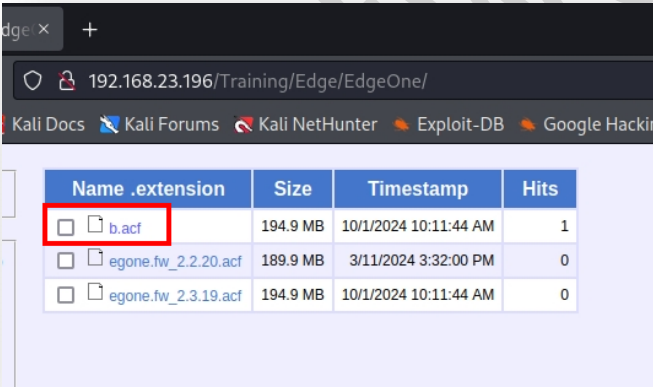
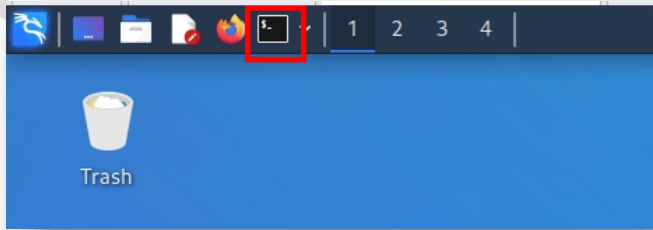
Items	Descriptions
	5. Allow the tool to communicate on both Public and Private networks, then click Allow Access.  6. Log in to the Win7 VM. 7. Open the Command Prompt. 8. Ping the following domain names. - www.txone.com - support.txone.com - www.ma.com - remote.ma.com 9. Go to the LabConsole VM and verify the ApateDNS console. To which domain name query was responded by ApateDNS? 

Items	Descriptions
	10. Navigate to the EdgeOne web UI 11. Go to Log & Report > Security Logs > Domain Name Filter Logs 12. Verify the logs for the domain name you queried in Step 8. Can you explain why the FQDN “support.txone.com” is allowed to respond, while the FQDN “remote.ma.com” is denied by EdgeIPS? 
Expected Result	Domain name profiles tested and working

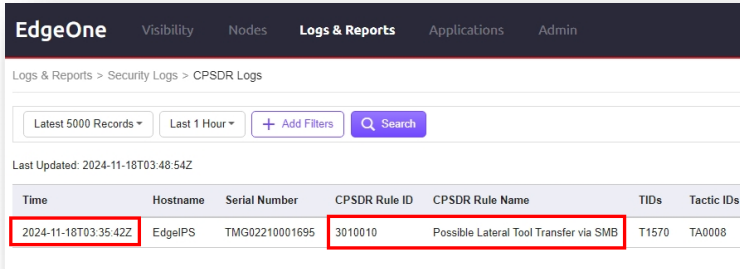
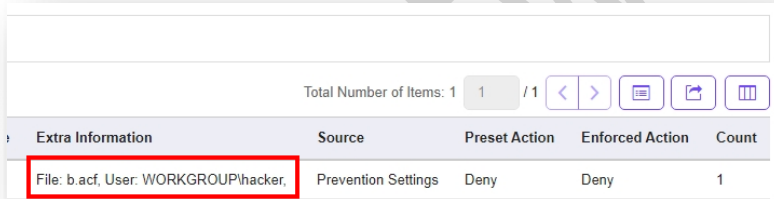
Lab7: CPSDR Networking

Table 10 CPSDR Networking

Items	Descriptions
Lab Case	CPSDR Networking, prevent the possible Lateral Tool Transfer via SMB
Steps	1. Log in to the LabConsole. 2. Navigate to the EdgeOne web UI. 3. Go to Node > EdgeIPS Family > EdgeIPS Std > Edge Group "FAB12-10-27" > CPSDR Threshold Settings. 4. Enable CPDR Detection Threshold. 5. Risk Level set to Medium and click Save.  6. Log in to Kali Linux, with the following credentials: - User: kali - Password: kali

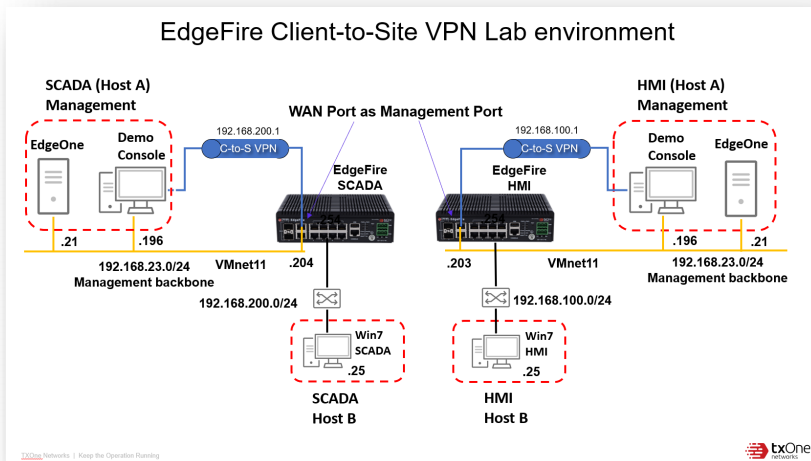
Items	Descriptions
	7. Start the Firefox browser  8. Go to http://192.168.23.196/Training/Edge/EdgeOne , download the file “b.acf” then save it to the Downloads folder.  9. Open the Terminal Emulator 

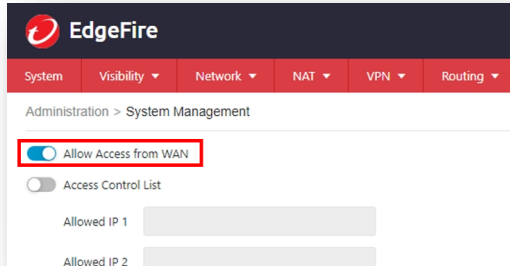
Items	Descriptions
10. Verify the file “b.acf” is in the Downloads folder.	<

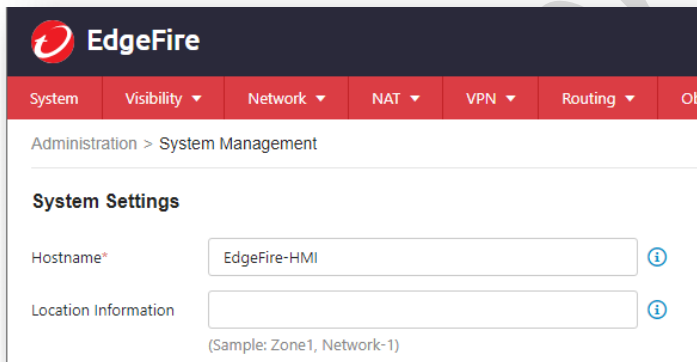
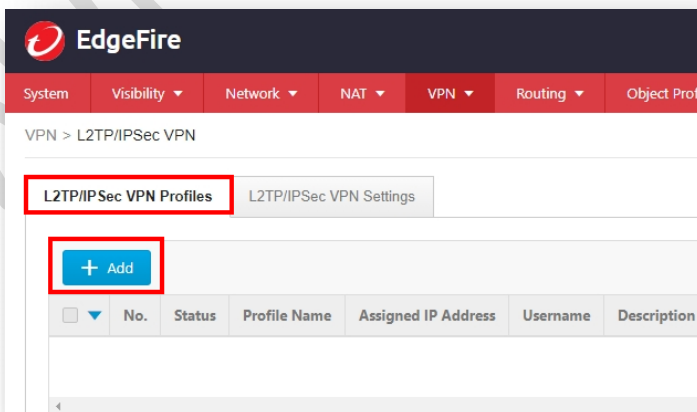
Items	Descriptions
	15. Log in to the LabConsole 16. Navigate to the EdgeOne web UI 17. Go to Logs & Reports > Security Logs > CPSDR Logs  18. Scroll to the Right, check the rest of the log-event 
Expected Result	The CPSDR engine has successfully blocked suspicious network behavior.

Lab8: EdgeFire Client-to-Site VPN Configuration

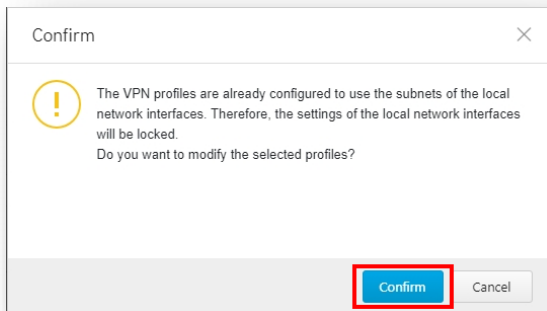
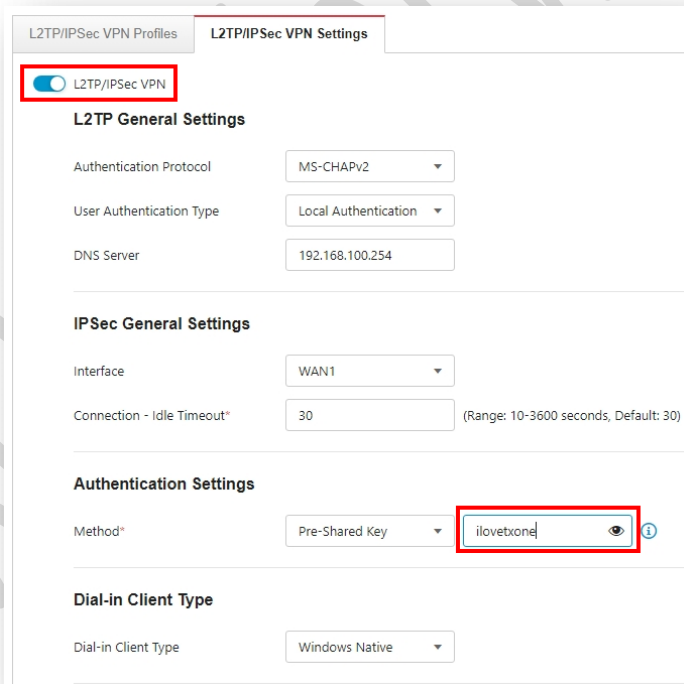
Table 11 EdgeFire Client-to-Site VPN Configuration

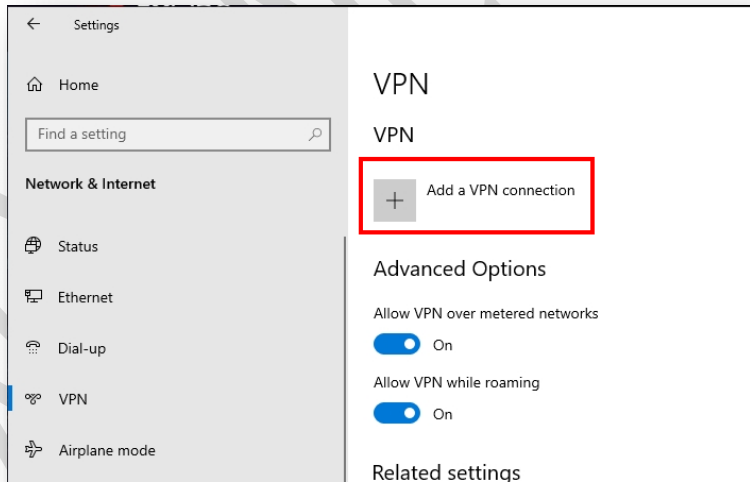
Items	Descriptions															
Lab Case	The EdgeFire Client-to-Site VPN Configuration															
	The EdgeFire Client-to-Site VPN Lab Environment  Prerequisites for the EdgeFire <table><tr><th>No</th><th>Tasks</th><th>Readiness</th></tr><tr><td>1</td><td>EdgeFire initialized and connected to EdgOne</td><td></td></tr><tr><td>2</td><td>EdgeFire firmware is v2 or higher</td><td></td></tr><tr><td>3</td><td>EdgeFire in the “FAB18-07-27” group in EdgeOne</td><td></td></tr><tr><td>4</td><td>EdgeOne Policy Enforcement set to Prevention Mode</td><td></td></tr></table> Refer to the Lab 9 in the 301 TXOne solutions basic implementation training lab guide for the configuration of the EdgeFire. The EdgeFire device IP address, EdgeOne IP address settings, please refer to the 401 Lab guide, and to Lab 7 EdgeFire Client- to-Site VPN Lab guide to configure the EdgeFire devices.	No	Tasks	Readiness	1	EdgeFire initialized and connected to EdgOne		2	EdgeFire firmware is v2 or higher		3	EdgeFire in the “ FAB18-07-27 ” group in EdgeOne		4	EdgeOne Policy Enforcement set to Prevention Mode	
No	Tasks	Readiness														
1	EdgeFire initialized and connected to EdgOne															
2	EdgeFire firmware is v2 or higher															
3	EdgeFire in the “ FAB18-07-27 ” group in EdgeOne															
4	EdgeOne Policy Enforcement set to Prevention Mode															

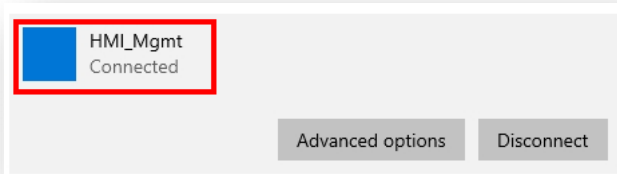
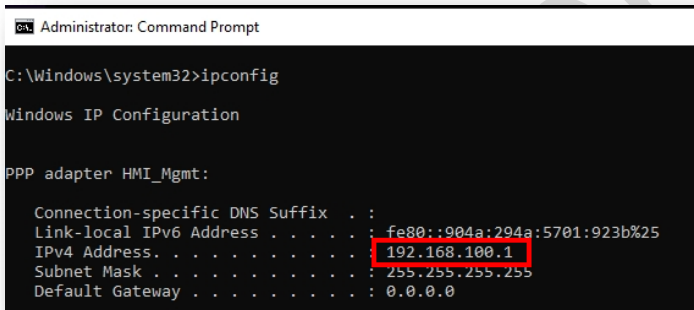
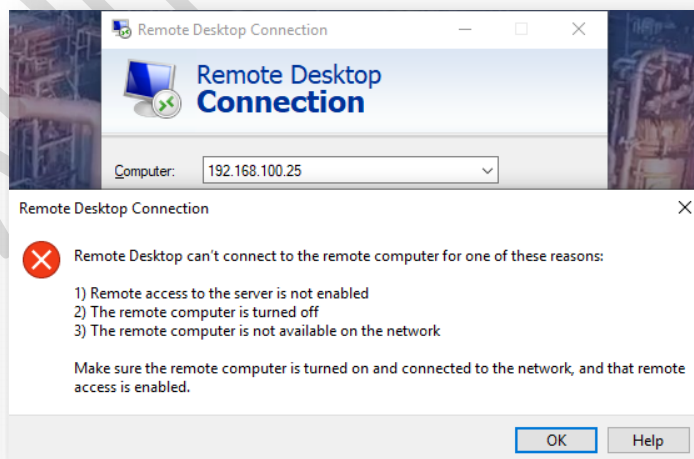
Items	Descriptions																																
	Note: Configure the LAN IP address and WAN IP address from the EdgeFire web UI: Administration > System Management and enable the option “Allow Access from WAN”  IP Address allocation table <table><tr><th>No.</th><th>VMs / Devices</th><th>HMI Networks</th><th>SCADA Networks</th></tr><tr><td>1</td><td>EdgeOne</td><td colspan="2">192.168.23.21</td></tr><tr><td>2</td><td>LabConsole</td><td colspan="2">192.168.23.196</td></tr><tr><td>3</td><td>EdgeFire Hostname</td><td>EdgeFire-HMI</td><td>EdgeFire-SCADA</td></tr><tr><td>4</td><td>EdgeFire WAN port</td><td>192.168.23.203</td><td>192.168.23.204</td></tr><tr><td>5</td><td>EdgeFire LAN port</td><td>192.168.100.254</td><td>192.168.200.254</td></tr><tr><td>6</td><td>Win7-HMI</td><td>Host Name: Win7-HMI IP: 192.168.100.25 GW: 192.168.100.254</td><td></td></tr><tr><td>7</td><td>Win7-SCADA</td><td></td><td>Host Name: Win7-SCADA IP: 192.168.200.25 GW: 192.168.200.254</td></tr></table> Note: - EdgeFire management will be on the WAN port	No.	VMs / Devices	HMI Networks	SCADA Networks	1	EdgeOne	192.168.23.21		2	LabConsole	192.168.23.196		3	EdgeFire Hostname	EdgeFire-HMI	EdgeFire-SCADA	4	EdgeFire WAN port	192.168.23.203	192.168.23.204	5	EdgeFire LAN port	192.168.100.254	192.168.200.254	6	Win7-HMI	Host Name: Win7-HMI IP: 192.168.100.25 GW: 192.168.100.254		7	Win7-SCADA		Host Name: Win7-SCADA IP: 192.168.200.25 GW: 192.168.200.254
No.	VMs / Devices	HMI Networks	SCADA Networks																														
1	EdgeOne	192.168.23.21																															
2	LabConsole	192.168.23.196																															
3	EdgeFire Hostname	EdgeFire-HMI	EdgeFire-SCADA																														
4	EdgeFire WAN port	192.168.23.203	192.168.23.204																														
5	EdgeFire LAN port	192.168.100.254	192.168.200.254																														
6	Win7-HMI	Host Name: Win7-HMI IP: 192.168.100.25 GW: 192.168.100.254																															
7	Win7-SCADA		Host Name: Win7-SCADA IP: 192.168.200.25 GW: 192.168.200.254																														

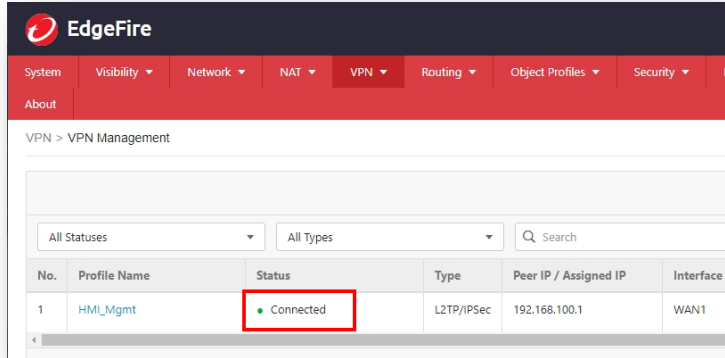
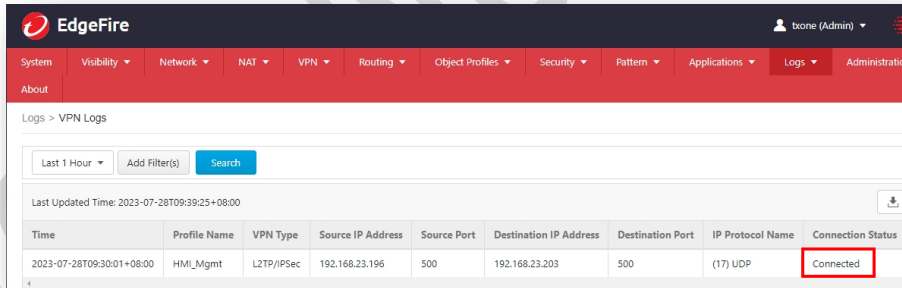
Items	Descriptions
	Configure the EdgeFire Client-to-Site VPN based on the table above. The screenshot uses the HMI networks as a reference - Log in to EdgeFire device web UI, with the following credentials - User: txone - Password: 12345678 - Go to Administration > System Management - Set the EdgeFire hostname to the EdgeFire-HMI or EdgeFire-SCADA  The screenshot shows the EdgeFire web interface. The top navigation bar includes System, Visibility, Network, NAT, VPN, Routing, and Object Profiles. The breadcrumb trail is Administration > System Management. Under System Settings, the Hostname field is set to EdgeFire-HMI and the Location Information field is empty. A sample text '(Sample: Zone1, Network-1)' is shown below the Location Information field. - Go to VPN > L2TP/IPSec VPN - Select the TAB L2TP/IPSec VPN Profiles and click +Add.  The screenshot shows the EdgeFire web interface for L2TP/IPSec VPN. The top navigation bar includes System, Visibility, Network, NAT, VPN, Routing, and Object Profiles. The breadcrumb trail is VPN > L2TP/IPSec VPN. There are two tabs: L2TP/IPSec VPN Profiles (selected) and L2TP/IPSec VPN Settings. In the L2TP/IPSec VPN Profiles tab, there is a blue '+ Add' button. Below the button is a table with columns: No., Status, Profile Name, Assigned IP Address, Username, and Description.

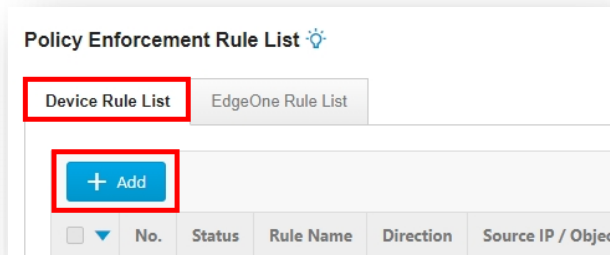
Items	Descriptions																								
6.	Configure the L2TP/IPSec VPN Profile information based on the table below. Edit L2TP/IPSec VPN Profile Status Profile Name* HMI_Mgmt Description Assigned IP Settings Assigned IP Address* 192.168.100.1 Interface LAN1 User Account Settings Username* hmi Password* ***** OK Cancel L2TP/IPSEC VPN Profiles summary table <table><tr><th>No.</th><th>Profile</th><th>HMI Network</th><th>SCADA Network</th></tr><tr><td>1</td><td>Profile Name</td><td>HMI_Mgmt</td><td>SCADA_Mgmt</td></tr><tr><td>2</td><td>Assigned IP Address</td><td>192.168.100.1</td><td>192.168.200.1</td></tr><tr><td>3</td><td>Interface</td><td colspan="2">LAN1</td></tr><tr><td>4</td><td>Username</td><td>hmi</td><td>scada</td></tr><tr><td>5</td><td>Password</td><td colspan="2">12345678</td></tr></table> 7. Click Save. EdgeFire System Visibility Network NAT VPN Routing Object Profiles Security VPN > L2TP/IPSec VPN L2TP/IPSec VPN Profiles L2TP/IPSec VPN Settings + Add No. Status Profile Name Assigned IP Address Username Description 1 HMI_Mgmt 192.168.100.1 hmi Save Cancel	No.	Profile	HMI Network	SCADA Network	1	Profile Name	HMI_Mgmt	SCADA_Mgmt	2	Assigned IP Address	192.168.100.1	192.168.200.1	3	Interface	LAN1		4	Username	hmi	scada	5	Password	12345678	
No.	Profile	HMI Network	SCADA Network																						
1	Profile Name	HMI_Mgmt	SCADA_Mgmt																						
2	Assigned IP Address	192.168.100.1	192.168.200.1																						
3	Interface	LAN1																							
4	Username	hmi	scada																						
5	Password	12345678																							

Items	Descriptions
	8. Click Confirm.  The VPN profiles are already configured to use the subnets of the local network interfaces. Therefore, the settings of the local network interfaces will be locked. Do you want to modify the selected profiles? Confirm Cancel 9. Select the L2TP/IPSec VPN Settings tab. 10. Enable the L2TP/IPSec VPN. 11. For the VPN settings, please refer to the summary table below.  L2TP/IPSec VPN Profiles L2TP/IPSec VPN Settings ☒ L2TP/IPSec VPN L2TP General Settings Authentication Protocol MS-CHAPv2 User Authentication Type Local Authentication DNS Server 192.168.100.254 IPSec General Settings Interface WAN1 Connection - Idle Timeout* 30 (Range: 10-3600 seconds, Default: 30) Authentication Settings Method* Pre-Shared Key ilovetxone Dial-in Client Type Dial-in Client Type Windows Native

Items	Descriptions																																
	The L2TP/IPSEC VPN Settings summary table <table><tr><th>No.</th><th>Settings</th><th>HMI Networks</th><th>SCADA Networks</th></tr><tr><td>1</td><td>Authentication Protocol</td><td colspan="2">MS-CHAPv2</td></tr><tr><td>2</td><td>User Authentication Type</td><td colspan="2">Local Authentication</td></tr><tr><td>3</td><td>DNS Server</td><td>192.168.100.254</td><td>192.168.200.254</td></tr><tr><td>4</td><td>IPSec Interface</td><td colspan="2">WAN1</td></tr><tr><td>5</td><td>Authentication Method</td><td colspan="2">Pre-Shared Key</td></tr><tr><td>6</td><td>Pre-Shared Key</td><td colspan="2">ilovetxone</td></tr><tr><td>7</td><td>Dial-in client</td><td colspan="2">Windows Native</td></tr></table> 12. Go to the windows Control Panel of the LabConsole VM. 13. Go to Network & Internet > VPN. 14. Click + Add a VPN connection. 	No.	Settings	HMI Networks	SCADA Networks	1	Authentication Protocol	MS-CHAPv2		2	User Authentication Type	Local Authentication		3	DNS Server	192.168.100.254	192.168.200.254	4	IPSec Interface	WAN1		5	Authentication Method	Pre-Shared Key		6	Pre-Shared Key	ilovetxone		7	Dial-in client	Windows Native	
No.	Settings	HMI Networks	SCADA Networks																														
1	Authentication Protocol	MS-CHAPv2																															
2	User Authentication Type	Local Authentication																															
3	DNS Server	192.168.100.254	192.168.200.254																														
4	IPSec Interface	WAN1																															
5	Authentication Method	Pre-Shared Key																															
6	Pre-Shared Key	ilovetxone																															
7	Dial-in client	Windows Native																															

Items	Descriptions
	18. Windows will reply that the VPN connection is established.  19. Open a CMD command prompt with Administrator rights. 20. Run IPCONFIG to verify the VPN network IP address.  21. Attempt to RDP to the HMI workstation at 192.168.100.25, but find that it is not possible. 

Items	Descriptions
	22. Go to the EdgeFire device web UI. 23. Go to VPN > VPN Management. 24. Verify the VPM status. The VPN connection is established.  25. Go to Logs > VPN Logs. 26. There should be a VPN access log-event from source IP 192.168.23.196.  27. Go to Security > Policy Enforcement. 28. The RDP access failed because there is no policy enforcement rule to allow RDP access. Create a Policy Enforcement rule to allow RDP access from VPN to the HMI workstation 192.168.100.25. 29. Select the TAB Device Rule List.

Items	Descriptions																																				
30.	Click + Add. 																																				
31.	Configure the Policy Enforcement Rule to allow RDP access from VPN to LAN based on the table and screenshot below Policy Enforcement Rule summary table <table><tr><th>No.</th><th>Policy Enforcement settings</th><th>HMI Networks</th><th>SCADA Networks</th></tr><tr><td>1</td><td>Rule Name</td><td>Allow_RDP_Access_HMI</td><td>Allow_RDP_Access_SCADA</td></tr><tr><td>2</td><td>Direction</td><td colspan="2">VPN to LAN</td></tr><tr><td>3</td><td>Source IP / Single IP</td><td>192.168.100.1</td><td>192.168.200.1</td></tr><tr><td>4</td><td>Destination IP / Single IP</td><td>192.168.100.25</td><td>192.168.200.25</td></tr><tr><td>5</td><td>Service Object</td><td colspan="2">TCP</td></tr><tr><td>6</td><td>Protocol Number</td><td colspan="2">6</td></tr><tr><td>7</td><td>Service Ports</td><td colspan="2">Custom : 3389~3389</td></tr><tr><td>8</td><td>Action</td><td colspan="2">Accept and Log</td></tr></table>	No.	Policy Enforcement settings	HMI Networks	SCADA Networks	1	Rule Name	Allow_RDP_Access_HMI	Allow_RDP_Access_SCADA	2	Direction	VPN to LAN		3	Source IP / Single IP	192.168.100.1	192.168.200.1	4	Destination IP / Single IP	192.168.100.25	192.168.200.25	5	Service Object	TCP		6	Protocol Number	6		7	Service Ports	Custom : 3389~3389		8	Action	Accept and Log	
No.	Policy Enforcement settings	HMI Networks	SCADA Networks																																		
1	Rule Name	Allow_RDP_Access_HMI	Allow_RDP_Access_SCADA																																		
2	Direction	VPN to LAN																																			
3	Source IP / Single IP	192.168.100.1	192.168.200.1																																		
4	Destination IP / Single IP	192.168.100.25	192.168.200.25																																		
5	Service Object	TCP																																			
6	Protocol Number	6																																			
7	Service Ports	Custom : 3389~3389																																			
8	Action	Accept and Log																																			

Items

Descriptions

Create Policy Enforcement Rule

Status

Rule Name*

Allow_RDP_Access_HMI

Description

Basic Filter

Direction

VPN to LAN

Source IP / Object

Single IP

192.168.100.1

Destination IP / Object

Single IP

192.168.100.25

Service Object

TCP

Protocol Number

6

Service Ports

Custom

3389

3389

Action

Accept and Log

32. Click **OK**.

33. Click **Save** to enforce the policy.

Policy Enforcement Rule List

Device Rule List

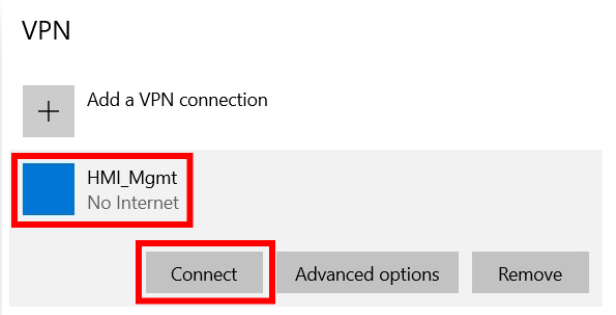
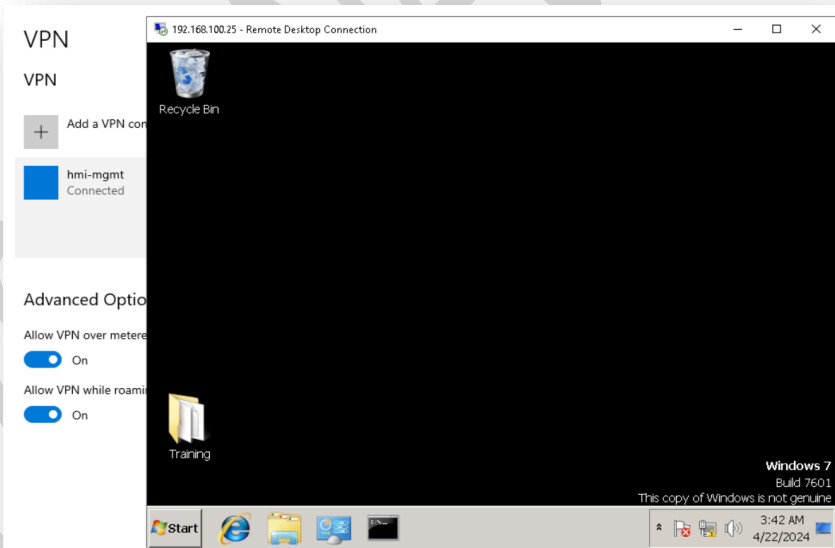
EdgeOne Rule List

+ Add

☐	No.	Status	Rule Name	Direction	Source IP / Object
☐	1		Allow_RDP_Access_HMI	VPN to LAN	User Defined

Save

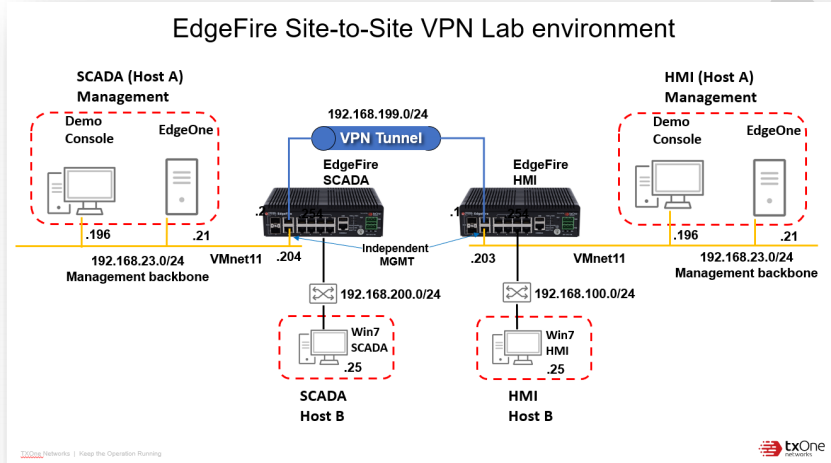
Cancel

Items	Descriptions
	34. Check the VPN connection status in Windows. If the VPN connection times out and disconnects, go to the Window Control Panel > Network & Interfaces > VPN and re-connect the VPN profile HMI_Mgmt / SCADA_Mgmt again.  35. RDP to the HMI / SCADA workstation 192.168.100.25 / 192.168.200.25 again. It should now be successful. 

Items	Descriptions																																				
	36. Go to the EdgeFire device web UI. 37. Go to Logs > Policy Enforcement Logs. 38. You should see “Allow” log-events for the connection to the HMI workstation.																																				
	EdgeFire System Viability Network NAT VPN Routing Object Profiles Security Pattern Applications Logs Administration About Logs > Policy Enforcement Logs Last 1 Hour Add Filter(s) Search Last Updated Time: 2023-07-28T10:18:02+08:00 <table><thead><tr><th>Time</th><th>Rule Name</th><th>Direction</th><th>Interface</th><th>Source IP Address</th><th>Source Port</th><th>Destination IP Address</th><th>Destination Port</th><th>Action</th></tr></thead><tbody><tr><td>2023-07-28T10:13:03+08:00</td><td>DeviceAllow_RDP_Access_HMI</td><td>VPN to LAN</td><td>L2TP/IPSEC/HMI_Mgmt</td><td>192.168.100.1</td><td>49708</td><td>192.168.100.25</td><td>3389</td><td>Allow</td></tr><tr><td>2023-07-28T10:12:59+08:00</td><td>DeviceAllow_RDP_Access_HMI</td><td>VPN to LAN</td><td>L2TP/IPSEC/HMI_Mgmt</td><td>192.168.100.1</td><td>49707</td><td>192.168.100.25</td><td>3389</td><td>Allow</td></tr><tr><td>2023-07-28T10:12:44+08:00</td><td>DeviceAllow_RDP_Access_HMI</td><td>VPN to LAN</td><td>L2TP/IPSEC/HMI_Mgmt</td><td>192.168.100.1</td><td>49706</td><td>192.168.100.25</td><td>3389</td><td>Allow</td></tr></tbody></table>	Time	Rule Name	Direction	Interface	Source IP Address	Source Port	Destination IP Address	Destination Port	Action	2023-07-28T10:13:03+08:00	DeviceAllow_RDP_Access_HMI	VPN to LAN	L2TP/IPSEC/HMI_Mgmt	192.168.100.1	49708	192.168.100.25	3389	Allow	2023-07-28T10:12:59+08:00	DeviceAllow_RDP_Access_HMI	VPN to LAN	L2TP/IPSEC/HMI_Mgmt	192.168.100.1	49707	192.168.100.25	3389	Allow	2023-07-28T10:12:44+08:00	DeviceAllow_RDP_Access_HMI	VPN to LAN	L2TP/IPSEC/HMI_Mgmt	192.168.100.1	49706	192.168.100.25	3389	Allow
Time	Rule Name	Direction	Interface	Source IP Address	Source Port	Destination IP Address	Destination Port	Action																													
2023-07-28T10:13:03+08:00	DeviceAllow_RDP_Access_HMI	VPN to LAN	L2TP/IPSEC/HMI_Mgmt	192.168.100.1	49708	192.168.100.25	3389	Allow																													
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2023-07-28T10:12:44+08:00	DeviceAllow_RDP_Access_HMI	VPN to LAN	L2TP/IPSEC/HMI_Mgmt	192.168.100.1	49706	192.168.100.25	3389	Allow																													
Expected Result	RDP from an “external” network to a private workstation successfully configured.																																				

Lab9: EdgeFire Site-to-Site VPN Configuration

Table 12 EdgeFire Site-to-Site VPN Configuration

Items	Descriptions																																			
Lab Case	The EdgeFire Site-to-Site VPN Configuration																																			
Steps	Lab Environment  HMI and SCADA Networks: <table><tr><th>IP Address type</th><th>HMI Network</th><th>SCADA Network</th></tr><tr><td>Connection Type</td><td colspan="2">Static IP</td></tr><tr><td>IP Address</td><td>192.168.199.1</td><td>192.168.199.2</td></tr><tr><td>Netmask</td><td>255.255.255.0</td><td>255.255.255.0</td></tr><tr><td>Gateway</td><td>192.168.199.2</td><td>192.168.199.1</td></tr></table> EdgeFire settings: <table><tr><th>No.</th><th>IP Address types</th><th>HMI Networks</th><th>SCADA Networks</th></tr><tr><td>1</td><td>EdgeOne</td><td colspan="2">192.168.23.21</td></tr><tr><td>2</td><td>LabConsole</td><td colspan="2">192.168.23.196</td></tr><tr><td>3</td><td>EdgeFire Hostname</td><td>EdgeFire-HMI</td><td>EdgeFire-SCADA</td></tr><tr><td>4</td><td>EdgeFire Management port</td><td>192.168.23.203</td><td>192.168.23.204</td></tr></table>	IP Address type	HMI Network	SCADA Network	Connection Type	Static IP		IP Address	192.168.199.1	192.168.199.2	Netmask	255.255.255.0	255.255.255.0	Gateway	192.168.199.2	192.168.199.1	No.	IP Address types	HMI Networks	SCADA Networks	1	EdgeOne	192.168.23.21		2	LabConsole	192.168.23.196		3	EdgeFire Hostname	EdgeFire-HMI	EdgeFire-SCADA	4	EdgeFire Management port	192.168.23.203	192.168.23.204
	IP Address type	HMI Network	SCADA Network																																	
	Connection Type	Static IP																																		
IP Address	192.168.199.1	192.168.199.2																																		
Netmask	255.255.255.0	255.255.255.0																																		
Gateway	192.168.199.2	192.168.199.1																																		
No.	IP Address types	HMI Networks	SCADA Networks																																	
1	EdgeOne	192.168.23.21																																		
2	LabConsole	192.168.23.196																																		
3	EdgeFire Hostname	EdgeFire-HMI	EdgeFire-SCADA																																	
4	EdgeFire Management port	192.168.23.203	192.168.23.204																																	

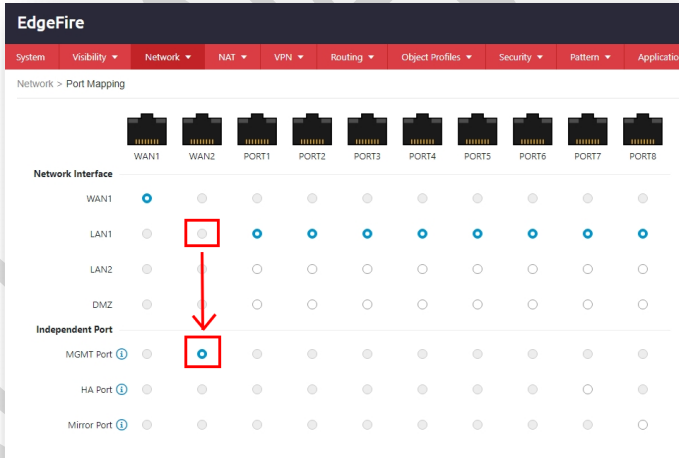
Items	Descriptions			
	5	EdgeFire WAN port	192.168.199.1	192.168.199.2
	6	EdgeFire WAN1 gateway	192.168.199.2	192.168.199.1
	7	EdgeFire LAN port	192.168.100.254	192.168.200.254
	8	Win7-HMI	192.168.100.25	
	9	Win7-SCADA		192.168.200.25

Note:
By default, EdgeFire is managed through its dedicated Management Port

EdgeFire Site-to-Site VPN configuration

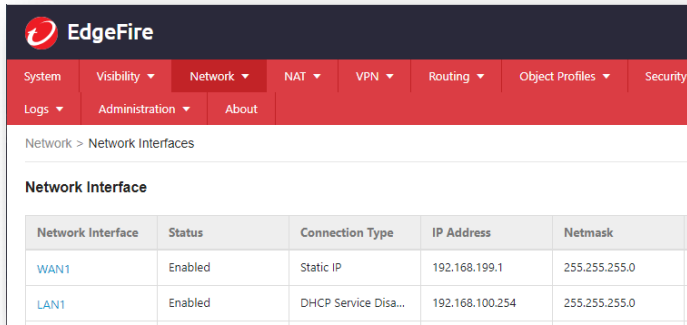
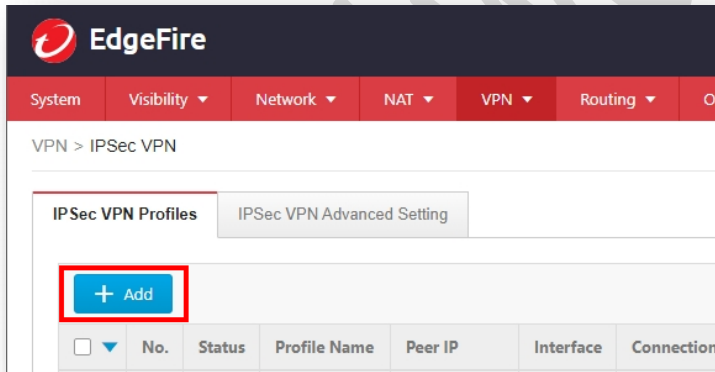
Change the management interface to WAN1

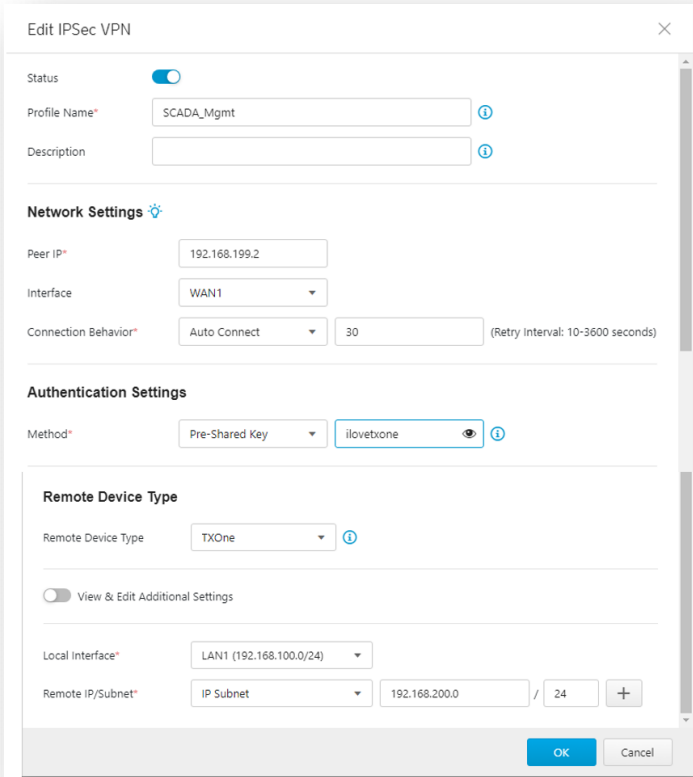
- Go to the EdgeFire device web UI > **Network Interface > Port Mapping**
- Set the WAN2 port to the **MGMT Port**

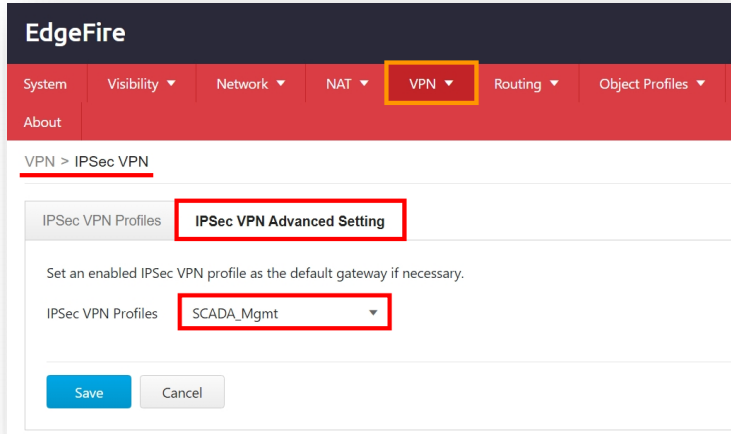
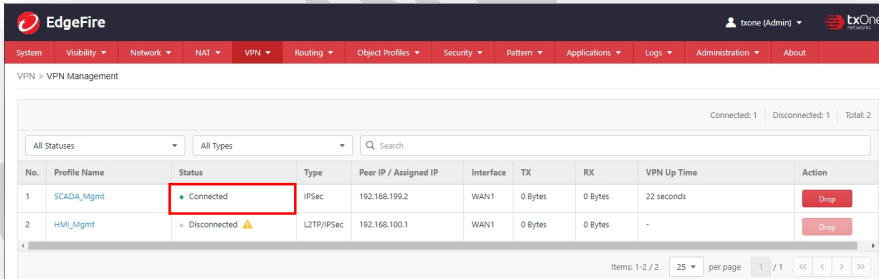


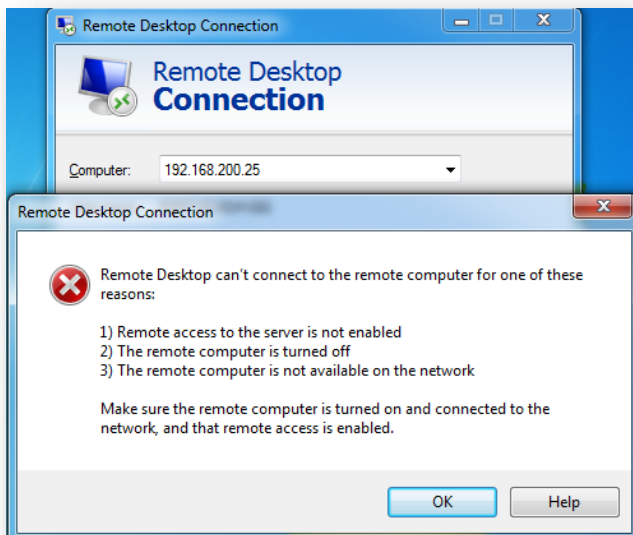
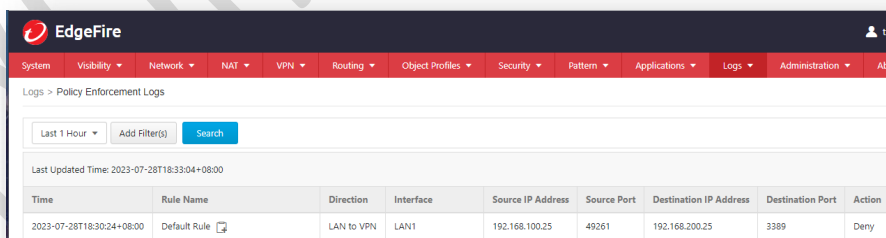
- Click **Save**.
- Go to **Network > Independent MGMT**.
- Configure the Management IP address with the same IP address as the WAN1 Port.

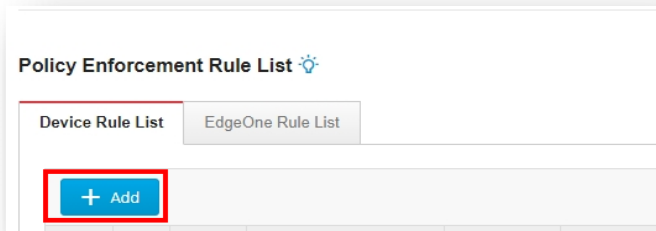
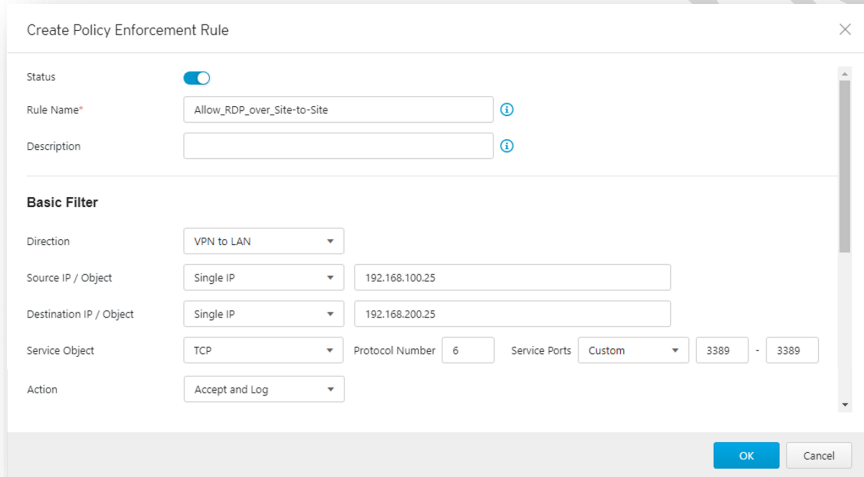
Items	Descriptions
	- Move the network cable from the WAN1 port to the Independent MGMT port. - Navigate to the EdgeFire device web UI. - Go to Network > Network Interfaces. - Configure the WAN1 Network Interface as indicated in the screenshot below.

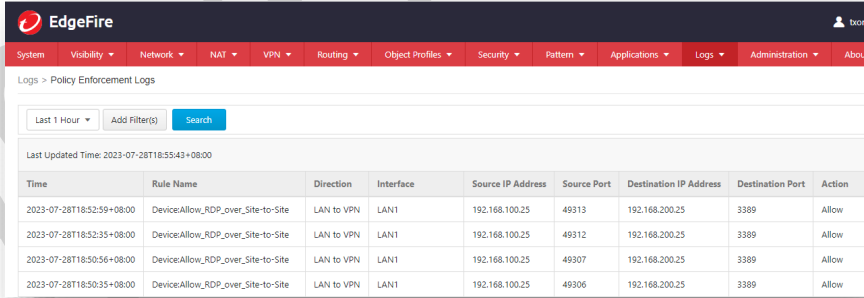
Items	Descriptions
	10. Verify that the WAN1 port has been bound to the right IP address.  11. Go to VPN > IPsec VPN > IPsec VPN Profiles. 12. Click +Add. 

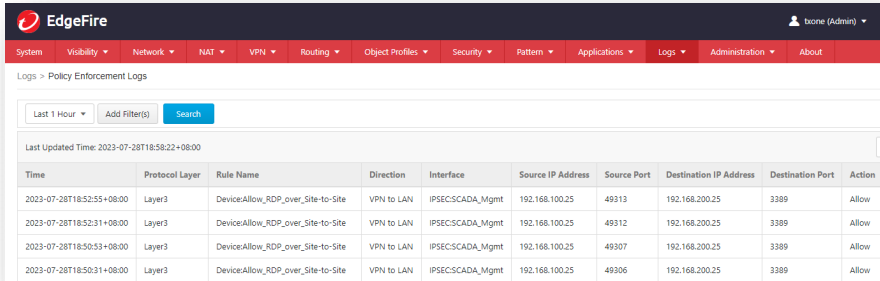
Items	Descriptions																																								
13.	Create an IPSec VPN based on the screenshot and table below.  IPSec VPN settings summary table <table><tr><th>No.</th><th>IPSec VPN settings</th><th>HMI Networks</th><th>SCADA Networks</th></tr><tr><td>1</td><td>Profile Name</td><td colspan="2">SCADA_Mgmt</td></tr><tr><td>2</td><td>Peer IP</td><td>192.168.199.2</td><td>192.168.199.1</td></tr><tr><td>3</td><td>Interface</td><td colspan="2">WAN1</td></tr><tr><td>4</td><td>Method</td><td colspan="2">Pre-Shared Key</td></tr><tr><td>5</td><td>Pre-Shared Key</td><td colspan="2">ilovetxone</td></tr><tr><td>6</td><td>Remote Device Type</td><td colspan="2">TXOne</td></tr><tr><td>7</td><td>Local Interface</td><td>LAN1(192.168.100.0/24)</td><td>LAN1(192.168.200.0)/24</td></tr><tr><td>8</td><td>Remote IP/Subnet</td><td colspan="2">IP Subnet</td></tr><tr><td>9</td><td>Subnet</td><td>192.168.200.0/24</td><td>192.168.100.0/24</td></tr></table>	No.	IPSec VPN settings	HMI Networks	SCADA Networks	1	Profile Name	SCADA_Mgmt		2	Peer IP	192.168.199.2	192.168.199.1	3	Interface	WAN1		4	Method	Pre-Shared Key		5	Pre-Shared Key	ilovetxone		6	Remote Device Type	TXOne		7	Local Interface	LAN1(192.168.100.0/24)	LAN1(192.168.200.0)/24	8	Remote IP/Subnet	IP Subnet		9	Subnet	192.168.200.0/24	192.168.100.0/24
No.	IPSec VPN settings	HMI Networks	SCADA Networks																																						
1	Profile Name	SCADA_Mgmt																																							
2	Peer IP	192.168.199.2	192.168.199.1																																						
3	Interface	WAN1																																							
4	Method	Pre-Shared Key																																							
5	Pre-Shared Key	ilovetxone																																							
6	Remote Device Type	TXOne																																							
7	Local Interface	LAN1(192.168.100.0/24)	LAN1(192.168.200.0)/24																																						
8	Remote IP/Subnet	IP Subnet																																							
9	Subnet	192.168.200.0/24	192.168.100.0/24																																						

Items	Descriptions
	14. Go to VPN > IPsec VPN > IPsec VPN Advanced Settings. Select the IPsec VPN Profiles: SCADA_Mgmt / HMI_Mgmt  15. Connect the network cable to the WAN1 port for both EdgeFire devices. The IPsec VPN should be activated automatically. 

Items	Descriptions
	16. Log in to the Win7-HMI. 17. Run mstsc.exe and RDP to the Win7-SCADA (192.168.200.25). It should still fail.  18. Navigate to the EdgeFire-HMI device web UI. 19. Go to Logs > Policy Enforcement Logs. 20. The web UI displays the deny logs. Verify the deny logs information, direction, source IP address, destination IP address and destination port.  21. Configure the Policy Enforcement rules to allow RDP traffic pass through. 22. Go to Security > Policy Enforcement.

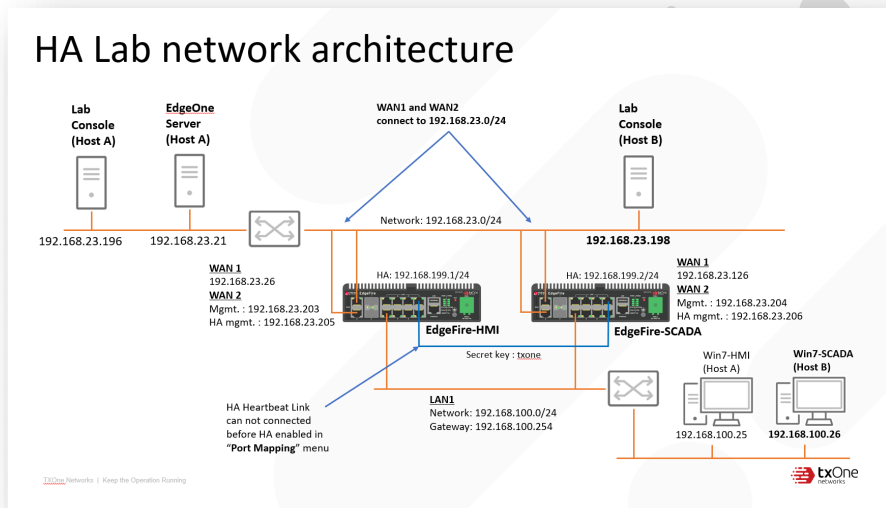
Items	Descriptions																																				
23.	Go to the Device Rule List tab and click +Add .																																				
																																					
24.	Configure the RDP access rules based on the table below.																																				
																																					
	The Policy Enforcement Rule settings summary table <table><tr><th>No.</th><th>Policy Enforcement settings</th><th>HMI Networks</th><th>SCADA Networks</th></tr><tr><td>1</td><td>Rule Name</td><td colspan="2">Allow_RDP_over_Site-to-Site</td></tr><tr><td>2</td><td>Direction</td><td>LAN to VPN</td><td>VPN to LAN</td></tr><tr><td>3</td><td>Source IP / Single IP</td><td colspan="2">192.168.100.25</td></tr><tr><td>4</td><td>Destination IP / Single IP</td><td colspan="2">192.168.200.25</td></tr><tr><td>5</td><td>Service Object</td><td colspan="2">TCP</td></tr><tr><td>6</td><td>Protocol Number</td><td colspan="2">6</td></tr><tr><td>7</td><td>Service Ports</td><td colspan="2">Custom: 3389~3389</td></tr><tr><td>8</td><td>Action</td><td colspan="2">Accept and Log</td></tr></table>	No.	Policy Enforcement settings	HMI Networks	SCADA Networks	1	Rule Name	Allow_RDP_over_Site-to-Site		2	Direction	LAN to VPN	VPN to LAN	3	Source IP / Single IP	192.168.100.25		4	Destination IP / Single IP	192.168.200.25		5	Service Object	TCP		6	Protocol Number	6		7	Service Ports	Custom: 3389~3389		8	Action	Accept and Log	
No.	Policy Enforcement settings	HMI Networks	SCADA Networks																																		
1	Rule Name	Allow_RDP_over_Site-to-Site																																			
2	Direction	LAN to VPN	VPN to LAN																																		
3	Source IP / Single IP	192.168.100.25																																			
4	Destination IP / Single IP	192.168.200.25																																			
5	Service Object	TCP																																			
6	Protocol Number	6																																			
7	Service Ports	Custom: 3389~3389																																			
8	Action	Accept and Log																																			

Items	Descriptions
	25. Click OK. 26. RDP from the Win7-HMI to Win7-SCADA again. It should now work.  27. Navigate to the EdgeFire-HMI device web UI or EdgeOne web UI to check the Policy Enforcement logs. 

Items	Descriptions
	28. Navigate to the EdgeFire-SCADA device web UI or EdgeOne web UI to check the Policy Enforcement logs. 
Expected Result	RDP access from the HMI workstation to the SCADA workstation via the EdgeFire Site-to-Site VPN successfully configured

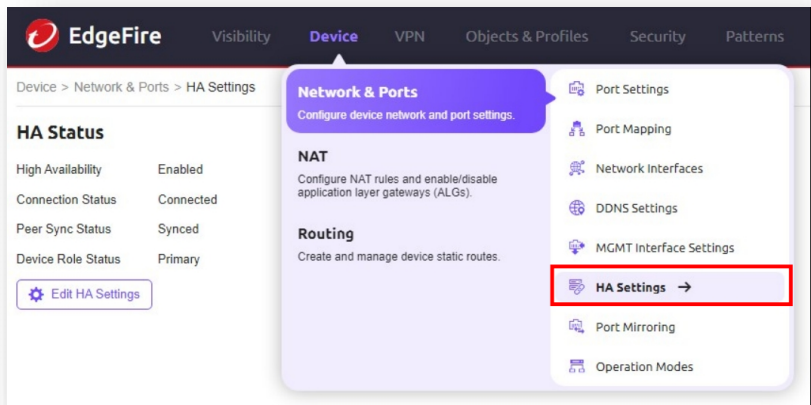
Lab10: EdgeFire High availability Configuration

Table 13 EdgeFire High availability Configuration

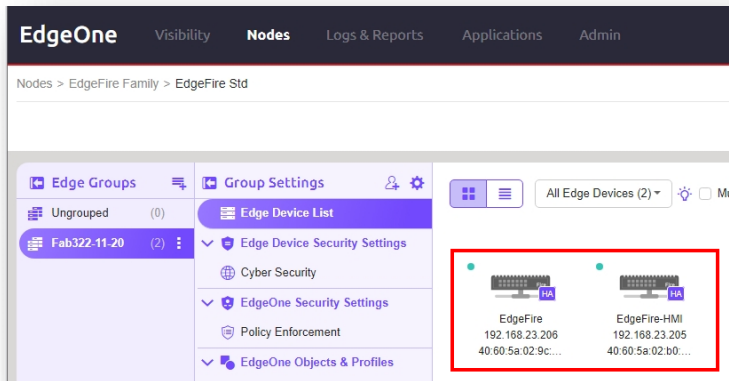
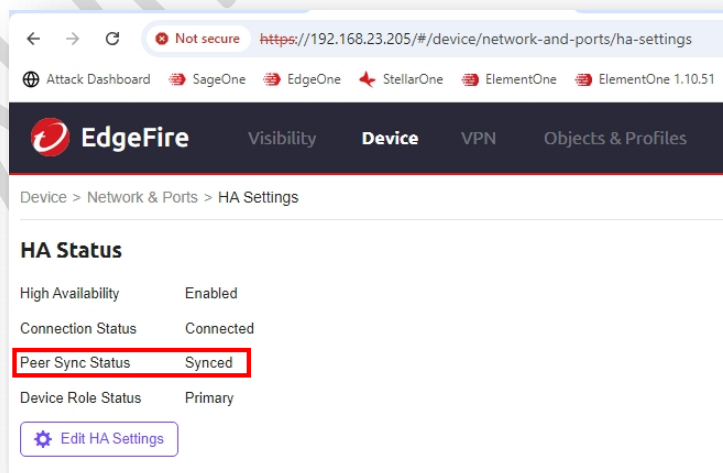
Items	Descriptions																																				
Lab Case	The EdgeFire High Availability Configuration																																				
	The EdgeFire High Availability Lab Environment HA Lab network architecture 																																				
Steps	The IP Address allocation table for the EdgeFire HA Lab pre-configurations <table><tr><th>No.</th><th>IP Address types</th><th>HMI Networks</th><th>SCADA Networks</th></tr><tr><td>1</td><td>EdgeOne</td><td colspan="2">192.168.23.21</td></tr><tr><td>2</td><td>LabConsole</td><td>192.168.23.196</td><td>192.168.23.198</td></tr><tr><td>3</td><td>EdgeFire Hostname</td><td>EdgeFire-HMI</td><td>EdgeFire-SCADA</td></tr><tr><td>4</td><td>EdgeFire Independent Management port</td><td>192.168.23.203</td><td>192.168.23.204</td></tr><tr><td>5</td><td>EdgeFire WAN port</td><td>192.168.23.26</td><td>192.168.23.126</td></tr><tr><td>7</td><td>EdgeFire LAN port</td><td>192.168.100.254</td><td>192.168.100.254</td></tr><tr><td>8</td><td>Win7-HMI</td><td>192.168.100.25</td><td></td></tr><tr><td>9</td><td>Win7-SCADA</td><td></td><td>192.168.100.26</td></tr></table> Note: EdgeFire management uses the Independent Management Port	No.	IP Address types	HMI Networks	SCADA Networks	1	EdgeOne	192.168.23.21		2	LabConsole	192.168.23.196	192.168.23.198	3	EdgeFire Hostname	EdgeFire-HMI	EdgeFire-SCADA	4	EdgeFire Independent Management port	192.168.23.203	192.168.23.204	5	EdgeFire WAN port	192.168.23.26	192.168.23.126	7	EdgeFire LAN port	192.168.100.254	192.168.100.254	8	Win7-HMI	192.168.100.25		9	Win7-SCADA		192.168.100.26
No.	IP Address types	HMI Networks	SCADA Networks																																		
1	EdgeOne	192.168.23.21																																			
2	LabConsole	192.168.23.196	192.168.23.198																																		
3	EdgeFire Hostname	EdgeFire-HMI	EdgeFire-SCADA																																		
4	EdgeFire Independent Management port	192.168.23.203	192.168.23.204																																		
5	EdgeFire WAN port	192.168.23.26	192.168.23.126																																		
7	EdgeFire LAN port	192.168.100.254	192.168.100.254																																		
8	Win7-HMI	192.168.100.25																																			
9	Win7-SCADA		192.168.100.26																																		

Items	Descriptions
	Note: Never connect to the Heartbeat link (both EdgeFire port 7) before enabling HA port mapping. The LED on port 7 will keep flashing if the EdgeFire Heartbeat link is connected before HA is enabled in Port Mapping. Game rules of EdgeFire HA - Which EdgeFire will become the primary after HA synchronization? Answer: The device that was the first to be powered on. - Which EdgeFire's configuration will be synced to the others? Answer: The configuration will be based on the EdgeFire where the 'Sync' button is pressed. Pre-Configuration tasks for EdgeFire HA Refer to the IP address allocation table for EdgeFire HA pre-configuration tasks to set up the EdgeFire devices - Shut down or disconnect the EdgeOne (SCADA) from the lab network. - Power off LabConsole (SCADA) and re-generate the Network Adapter MAC Address. - Power on LabConsole (SCADA) and set the IP address to 192.168.23.198/24. - Create an admin account to access the EdgeOne (HMI). - Username: txone-scada - Password: P@ssw0rd - Set the IP address for the EdgeFire WAN1 port.

Items	Descriptions
	6. Set the IP address for the EdgeFire management port. 7. Wire the following servers, endpoints, and devices to a network switch. - LabConsole (HMI) - EdgeOne Server (HMI) - LabConsole (SCADA) - EdgeFire-HMI: WAN1 port and management Port - EdgeFire-SCADA: WAN1 port and management Port 8. Register the EdgeFire-SCADA with the EdgeOne (HMI) and move it into the Edge Group "FAB18-07-27". 9. Power off Win7 (SCADA) and re-generate the Network Adapter MAC Address. 10. Power on Win7 (SCADA) and set the IP address to 192.168.100.26/24 and the gateway IP address point to 192.168.100.254. 11. Wire the following endpoints and devices to a network switch - Win7-HMI - Win7-SCADA - EdgeFire-HMI LAN port - EdgeFire-SCADA LAN port 12. Navigate to the EdgeFire-HMI / EdgeFire-SCADA device web UI. 13. Go to Device > Network & Ports > HA Settings.

Items	Descriptions																																
	 The IP Address allocation table for the EdgeFire HA Labs configurations <table><tr><th>No.</th><th>IP Address types</th><th>HMI Networks</th><th>SCADA Networks</th></tr><tr><td>1</td><td>HA Interface IP Address</td><td>192.168.199.1</td><td>192.168.199.2</td></tr><tr><td>2</td><td>HA Interface Subnet Mask</td><td>255.255.255.0</td><td>255.255.255.0</td></tr><tr><td>3</td><td>Unicast Heartbeat Peer IP</td><td>192.168.199.2</td><td>192.168.199.1</td></tr><tr><td>4</td><td>Shared Secret Key</td><td colspan="2">txone</td></tr><tr><td>5</td><td>Heartbeat Interval</td><td colspan="2">1 Second</td></tr><tr><td>7</td><td>Failover Trigger Level</td><td colspan="2">3 Times</td></tr><tr><td>9</td><td>Management IP for Independent MGMT</td><td>192.168.23.205</td><td>192.168.23.206</td></tr></table>	No.	IP Address types	HMI Networks	SCADA Networks	1	HA Interface IP Address	192.168.199.1	192.168.199.2	2	HA Interface Subnet Mask	255.255.255.0	255.255.255.0	3	Unicast Heartbeat Peer IP	192.168.199.2	192.168.199.1	4	Shared Secret Key	txone		5	Heartbeat Interval	1 Second		7	Failover Trigger Level	3 Times		9	Management IP for Independent MGMT	192.168.23.205	192.168.23.206
No.	IP Address types	HMI Networks	SCADA Networks																														
1	HA Interface IP Address	192.168.199.1	192.168.199.2																														
2	HA Interface Subnet Mask	255.255.255.0	255.255.255.0																														
3	Unicast Heartbeat Peer IP	192.168.199.2	192.168.199.1																														
4	Shared Secret Key	txone																															
5	Heartbeat Interval	1 Second																															
7	Failover Trigger Level	3 Times																															
9	Management IP for Independent MGMT	192.168.23.205	192.168.23.206																														

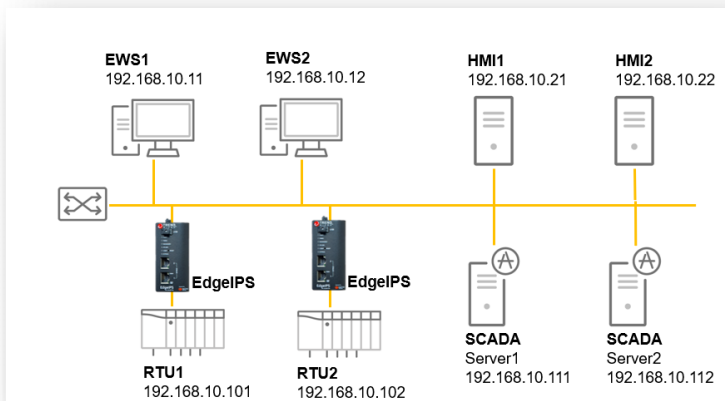
Items	Descriptions
	14. Set the HA Settings based on to the IP allocation table. 15. Verify that the High Availability status remains “Enabled”. 16. Connect the Heartbeat network cable to port 7 on both EdgeFire devices. 17. Navigate to the EdgeFire-HMI device web UI 18. Go to Device > Network & Ports > HA Settings 19. Verify that the connection status remains “Connected” and click Sync.

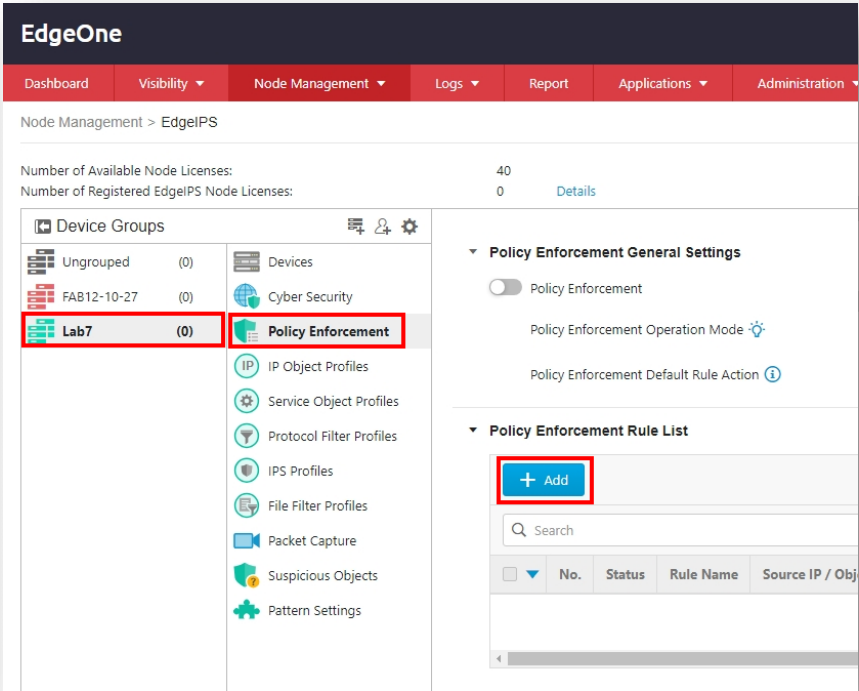
Items	Descriptions
	20. Navigate to the EdgeOne web UI. 21. Go to Node > EdgeFire Family > EdgeFire Std. 22. Verify that the EdgeFire status changes to the GREEN light and the HA icon color changes from RED to PURPLE.  23. Navigate to the EdgeFire-HMI device web UI using the Management IP for Independent MGMT of HA operations mode. 24. Go to Device > Network & Ports > HA Settings. 25. Verify that the Peer Sync Status is “Synced”. 

Items	Descriptions
	26. Configure the NAT rules and policy enforcement rules on the EdgeFire-HMI and EdgeOne to allow the following access criteria. Ask the trainer to check the results once the configuration is complete. - RDP from the LabConsole (HMI) to the Win7 (HMI) - RDP from the LabConsole (SCADA) to the Win7 (SCADA) - Go to the EdgeFire-SCADA to verify the settings had replicated from EdgeFire-HMI or not - Power off the EdgeFire-HMI and verify the RDP connection
Expected Result	The RDP access from the LabConsole to the Win7 workstation via the EdgeFire HA was successful

Lab11: Lab Quiz: Build the Policy Enforcement Rule list (Optional)

Table 14 Build the Policy Enforcement Rule list

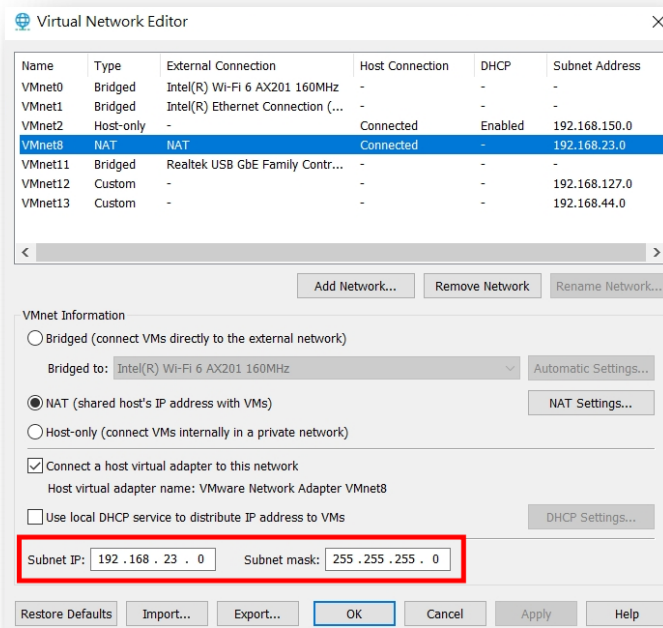
Items	Descriptions
Lab Case	Build Policy Enforcement Rule list
Steps	1. OT network architecture  2. OT operations workflow requirements A. EWS1 allow control access (READ/WRITE) to RTU1 over Modbus B. EWS2 allow control access (READ/WRITE) to RTU2 over Modbus C. Rest of others client in OT network that allow READ access to RTU1 over Modbus D. Rest of others client in OT network that allow READ access to RTU2 over Modbus

Items	Descriptions
	- Lab quiz Build the policy enforcement rules according to the “OT operations workflow requirements” from above. Please create rules under the Device Group – “LAB7” in EdgeOne
	 The screenshot displays the EdgeOne interface for managing policy enforcement. On the left, under 'Device Groups', 'Lab7 (0)' is selected. The right-hand menu highlights 'Policy Enforcement'. The main content area shows 'Policy Enforcement General Settings' with a toggle for 'Policy Enforcement' turned on. Below this, the 'Policy Enforcement Rule List' section features an 'Add' button and a table with headers: 'No.', 'Status', 'Rule Name', and 'Source IP / Obj'.
Expected Result	Policy enforcement rules created according to the “OT workflow requirements”

5. Appendix

5.1 How to get updates for TXOne in the Lab environment?

■ Configure VMWare NAT network to 192.168.23.0 /24



■ Switch Network Adapter to NAT for EdgeOne, LabConsole

- Default setting of NAT default gateway will point to 192.168.23.2
- LabConsole, EdgeOne have 8.8.8.8 as their name server.

